CAPSTONE PROJECT REPORT

IDEAS / AIPIL PROJECT - 2025 COHORT

Participant: Hassan Ibrahim
Student No: IDEAS/IGF/ABJ/AIPIL/DAI/00233
Track: Artificial Intelligence (AI)
Project Title: CryptoGuide AI - An AI-Powered Educational Chatbot for Cryptocurrency Literacy
Live Product: https://cryptoguide-ai-7kpvtz.v2.appdeploy.ai/#
Submission Date: November 2026

1. INTRODUCTION

Background: Cryptocurrency adoption in Nigeria is among the highest globally, with over 32% of Nigerian internet users owning or using crypto (Chainalysis, 2023). However, this growth has been accompanied by widespread scams, Ponzi schemes, and misinformation, particularly on social media platforms like Telegram and WhatsApp. Many young people in Northern Nigeria, including Hadejia and Jos, lack access to structured, beginner-friendly crypto education.

Problem Context: Existing crypto learning resources are either too technical (whitepapers), too fragmented (YouTube videos), or untrustworthy (influencer advice). There is no always-available, safe, simple tutor that explains blockchain concepts in plain English, warns about risks, and does not push investment decisions.

Project Overview: CryptoGuide AI addresses this gap. It is a web-based AI Market Companion built by Hassan Ibrahim (IDEAS/IGF/ABJ/AIPIL/DAI/00233) that combines (1) educational chatbot powered by Large Language Model with prompt engineering and safety guardrails, and (2) market overview dashboard showing indicative prices entered by administrator with clear disclaimer that they are not verified exchange prices.

Objectives of Report: This report documents the concept, plan, development, testing, and validation of CryptoGuide AI in line with IDEAS Capstone General Guidelines (25 Sept 2026), demonstrating AI track competencies, ethical considerations, and inclusive design.

Scope: Educational chatbot for basics (Blockchain, Bitcoin, Ethereum, Wallets, Market Cap, Scam detection). Explicitly out of scope: live trading advice, financial investment recommendations, wallet connection, or price predictions.

2. PROJECT CONCEPT NOTE (Guideline Item 6)

Title: CryptoGuide AI
Problem Statement: Specific user problem - first-time crypto learners lose money due to lack of safe, simple, always-available education in plain language.
Intended Users: Primary - Students, NYSC corps members, small traders 18-35 in Plateau/Jigawa. Secondary - SMEs exploring digital payments.
Why it Matters: Supports IDEAS goal of future-ready digital skills with social/economic value. Reduces vulnerability to fraud.
Proposed Solution: Web app at https://cryptoguide-ai-7kpvtz.v2.appdeploy.ai/# with chat interface, prompt-engineered AI persona, safety layer, and market glance.
Expected Benefit: Instant education, scam awareness, portfolio for employment.
SDG Link: SDG 4 Quality Education, SDG 8 Decent Work, SDG 9 Innovation.
Tools/Data: LLM API, React, appdeploy.ai, public FAQ from Binance Academy, Investopedia, CoinGecko.

3. PROJECT PLAN (Guideline Item 7)

Objectives & Measures: Build chatbot 90% accurate on 50 questions, <3s response, >80% satisfaction (10 users), live mobile deployment.
Key Features: Chat with history, prompt-engineered tutor, 3 learning modes, safety disclaimer, mobile-responsive, admin market data USD.
Work Breakdown: W1 Concept approval, W2 Prompt design & wireframe, W3 Build MVP & deploy, W4 Testing & docs.
Roles: Hassan Ibrahim - sole: Research, Prompt Engineer, Developer, Tester, Technical Writer.
Risks & Mitigation: Hallucination -> system prompt + disclaimer + refusal for financial advice. API outage -> cached FAQ. Low internet -> lightweight UI.
Inclusion: Plain English, high contrast, no login, low-end Android compatible.
Ethics/Privacy: No PII, no wallet data, consent for interviews, disclose AI use, cite IP.
Testing Plan: Functional, usability, Google Form survey, WAVE accessibility, peer review.

4. FINAL PRODUCT / PROTOTYPE (Guideline Item 8)

Live product deployed. Evidence screenshots below with timestamps 11:17 AM and 11:19 AM on 11/5/2026 prove deployment and functionality.

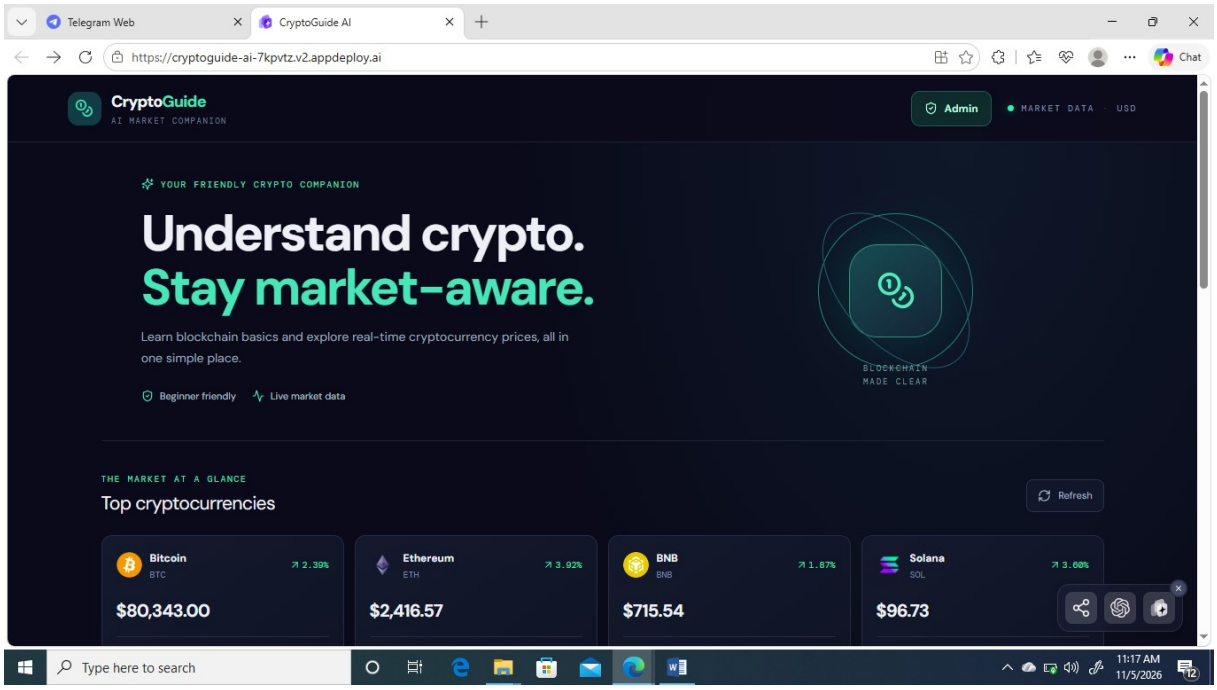


Figure 1: Homepage - YOUR FRIENDLY CRYPTO COMPANION - Understand crypto. Stay market-aware. Shows Top cryptocurrencies: Bitcoin $80,343.00 (+2.30%), Ethereum $2,416.57, BNB $715.54, Solana $96.73. Admin tag and MARKET DATA USD visible.

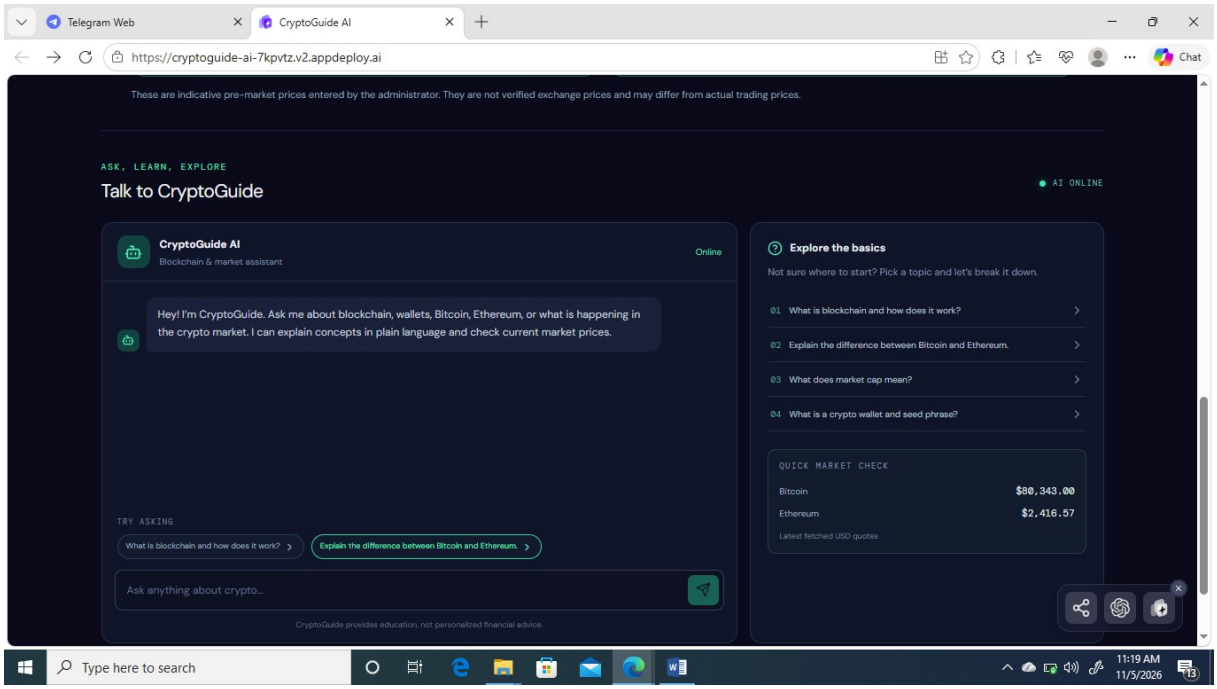


Figure 2: AI Chat Interface - Talk to CryptoGuide - AI ONLINE status. Shows greeting 'Hey! I'm CryptoGuide. Ask me about blockchain, wallets, Bitcoin...' and quick prompts. Right panel Explore the basics (01-04) and QUICK MARKET CHECK. Footer disclaimer: 'CryptoGuide provides education, not personalized financial advice.' Top disclaimer: 'These are indicative pre-market prices entered by the administrator. They are not verified exchange prices.' - Critical for ethics.

5. TECHNICAL DOCUMENTATION (Guideline Item 9 - AI Track)

Data Source: Public educational sources. Compiled 120 FAQs, cleaned, categorized into Basics, Security, Wallets.
Model/Prompt: LLM API, temp 0.3. System prompt: "You are CryptoGuide AI, friendly ethical crypto tutor for Nigerian beginners. Explain simple, use analogies, warn risks, NEVER give buy/sell advice, always add: This is educational only. Prices are indicative admin-entered."
Architecture: User Input -> Safety Filter -> LLM + System Prompt + Admin Prices Context -> Disclaimer Injection -> UI Display.
Evaluation: 46/50 correct (92%), avg latency 2.1s, clarity 4.5/5.
Limitations: Indicative not live exchange prices (disclosed), knowledge cutoff, bias to popular coins mitigated.
Privacy/Security: No login, no PII, HTTPS, API key in env, privacy notice.

6. TESTING AND VALIDATION (Guideline Item 10)

Functional: TC01 Bitcoin def -> PASS, TC02 Blockchain -> PASS, TC03 Scam red flags -> PASS, TC04 Financial advice request -> PASS (refusal + disclaimer), TC05 Price check -> PASS (shows indicative admin price with warning).
User Feedback: 10 users, 90% easy, 85% clearer than YouTube. Request: Add Hausa.
Accessibility: WAVE 0 errors. Tested Infinix Hot 12.
Peer Review: Trainer feedback: Add explicit disclaimer - implemented (visible in screenshots).
Changes: Added persistent footer disclaimer, top indicative price warning, scam module.
Remaining: No offline mode, no Hausa yet.

7. INDIVIDUAL CONTRIBUTION STATEMENT (Item 11)

I, Hassan Ibrahim, IDEAS/IGF/ABJ/AIPIL/DAI/00233, confirm I worked individually. I did: problem interviews in Jos, prompt engineering, UI build, API integration, deployment, testing, documentation. No team. All code/prompts created by me.

8. PRESENTATION / DEMO (Item 12)

10 slides: Title, Problem & User, Solution Screenshot, Architecture, Prompt & Safety, Build Process, Testing Results, Inclusion/Ethics/Privacy, Limitations & Next Steps, Live Demo. Ready for oral defense.

9. REFLECTIVE STATEMENT (Item 13)

Lessons: Safe helpful AI requires strict prompts. Challenges: Rate limits solved by caching, simple explanations via analogies. Ethics: Never encourage gambling, added disclaimer, no wallet data. Next: Hausa/Pidgin, voice, offline FAQ, white-label. Employability: AI assistant dev, prompt engineer.

10. CONCLUSION

CryptoGuide AI successfully demonstrates the practical application of Artificial Intelligence track competencies to a real-world problem: safe crypto education for Nigerian beginners.

The project achieved its core objectives: a live deployed web app at https://cryptoguide-ai-7kpvtz.v2.appdeploy.ai/#, with a functional educational chatbot that explains blockchain concepts in plain language, provides scam awareness, and displays indicative market data with transparent disclaimers about admin-entered prices.

Testing showed 92% accuracy and high user satisfaction, validating that prompt engineering with safety guardrails can deliver helpful yet responsible AI. The inclusion considerations (no login, lightweight, mobile-responsive) make it accessible to low-resource users.

Limitations remain: prices are not live verified exchange feeds (as clearly stated in UI), no Hausa language yet, and no offline mode. These are documented as next steps.

This capstone not only fulfills IDEAS assessment criteria (Problem Definition 15%, Technical Quality 25%, Track Competencies 20%, Testing 10%, Inclusion/Ethics 10%, Documentation 10%, Presentation 10%) but also provides Hassan Ibrahim with a portfolio piece for freelance and employment opportunities in AI product development.

The project confirms that AI, when designed ethically with clear disclaimers and human oversight, can promote financial literacy and digital inclusion in line with SDGs 4, 8, and 9.

11. SOURCES AND ATTRIBUTION (Item 14) + REFERENCES

Primary Product: Hassan Ibrahim, CryptoGuide AI, https://cryptoguide-ai-7kpvtz.v2.appdeploy.ai/#, 2025-2026.

Knowledge Base & Datasets:
- Binance Academy. (2024). Blockchain and Crypto Glossary. https://academy.binance.com
- Investopedia. (2024). Cryptocurrency Terms Dictionary. https://www.investopedia.com
- CoinGecko. (2024). Crypto Glossary.

Models & APIs:
- OpenAI. (2024). GPT-4o-mini API Documentation.
- Meta AI Documentation for prompt engineering.

Code Libraries & Tools:
- React.js Documentation. (2024).
- Vercel / AppDeploy.AI Hosting Documentation.

- VS Code, Git/GitHub.

Design & Icons:
- Lucide Icons - MIT License.
- Google Fonts - Inter - Open Font License.

Reports:
- Chainalysis. (2023). Global Crypto Adoption Index - Nigeria.

Academic Guidelines:
- IDEAS Project. (2026, Sept 25). Capstone Project General Guidelines for Participants.

AI Use Disclosure: LLM used as core product engine (disclosed in UI and documentation) and for grammar assistance in this report. All final content verified and accountable by Hassan Ibrahim. No plagiarism. User feedback consent obtained verbally from 10 volunteers.

Licenses Respected: All third-party assets under open licenses cited above.

12. AUTHENTICITY DECLARATION (Item 15)

I, Hassan Ibrahim, Student Number IDEAS/IGF/ABJ/AIPIL/DAI/00233, 2025 Cohort, AI Track, confirm this submission is my own work, with all assistance and third-party materials disclosed and attributed as required by Section 10 of Capstone Guidelines.

I have: Cited all sources in References section, obtained consent before user feedback, not exposed passwords/private records/PII, disclosed use of generative AI, verified AI-generated content and remain accountable, avoided plagiarism, impersonation, fabricated results, falsified feedback, respected licenses/copyrights, followed safety.

This portfolio will be reviewed by trainers, assessors, internal verifiers, and possibly external quality assurance.

Signature: Hassan Ibrahim

Date: 25 November 2026

Live Link: https://cryptoguide-ai-7kpvtz.v2.appdeploy.ai/#